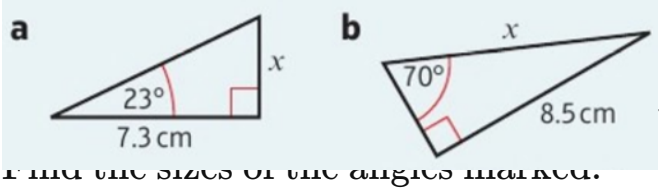
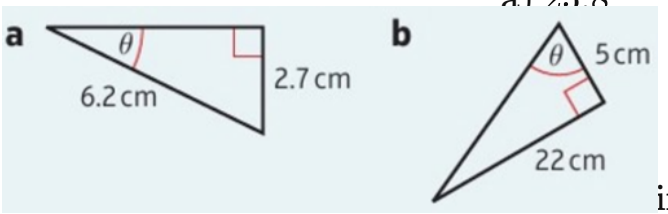
Y1 Chapter 9 Trigonometric Ratios

1) Use trigonometry to find the lengths of the marked sides.



2) Find the sizes of the angles marked.



3) intersection at:

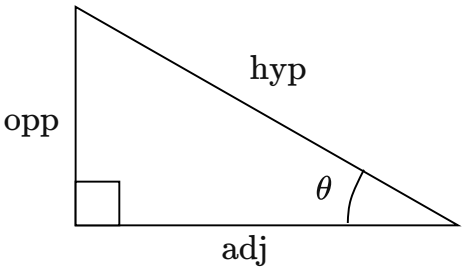
a) $y = f(x)$ — $(-3, 0), (0, 0)$
b) $y = f(x + 2)$ — $(-5, 0), (-2, 0), (0, 10)$
c) $y = f(x) - 3$ — $\left(\frac{-3 \pm \sqrt{21}}{2}\right), (0, -3)$
d) $y = f\left(\frac{1}{2}x\right)$ — $(-6, 0), (0, 0)$

a) $25.8°$

1: Sine/Cosine Rule

2: Areas of Triangles

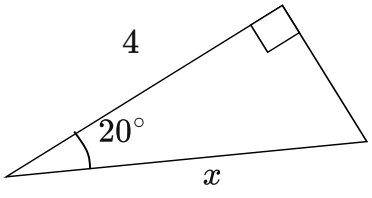
3: Graphs of Sine/Cosine/Tangent



You are probably familiar with the formula: $\sin(\theta) = \frac{opp}{hyp}$
But what is the *conceptual* definition of sin ?

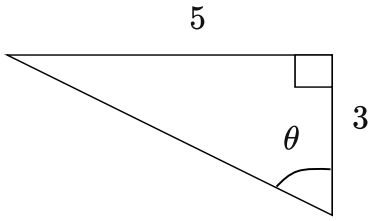
e

Remember that a ratio just means the 'relative size' between quantities (in this case lengths). For this reason, sin/cos/tan are known as "trigonometric ratios".



Find x.

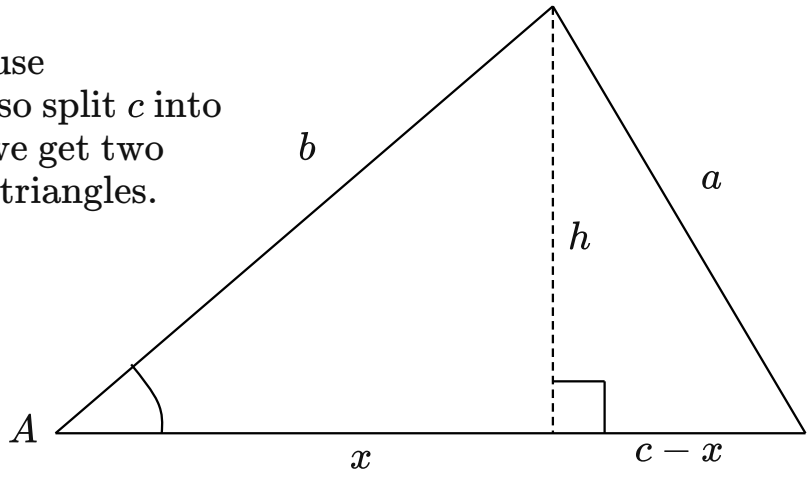
Tip: You can swap the thing you're dividing by and the result. e.g. $\frac{8}{2} = 4 \rightarrow \frac{8}{4} = 2$.



Find θ.

reflections: You may have been taught "use tan^{-1} whenever you're finding an angle", and therefore write the second line directly. It is preferable to always write the first line, then see the problem as a 'changing the subject' one. We need to remove the tan on front of the θ, so apply tan^{-1} to each side of the equation to 'cancel out' the tan on the LHS.

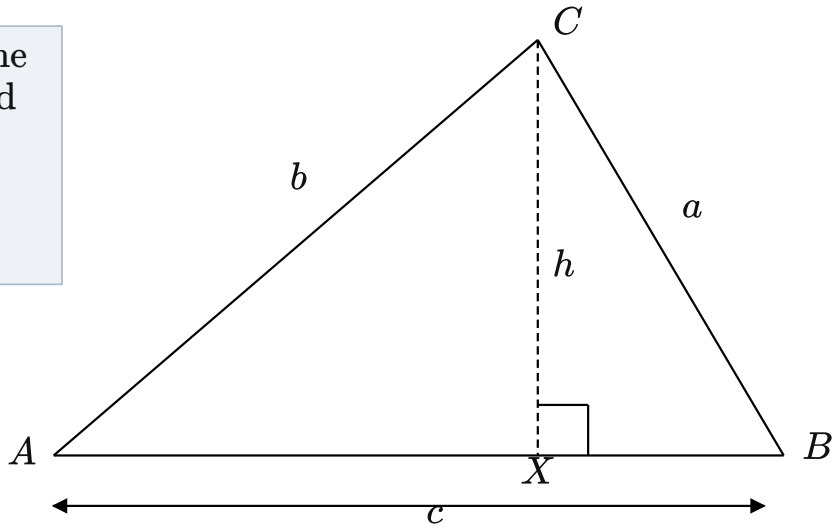
We want to use Pythagoras, so split c into two so that we get two right-angled triangles.



By Pythagoras: $h^2 + x^2 = b^2$ and $h^2 + (c-x)^2 = a^2$
Subtracting to eliminate h:
$x^2 - (c-x)^2 = b^2 - a^2$
$a^2 = b^2 + c^2 - 2cx$
But $x = b\cos A$ $\therefore$

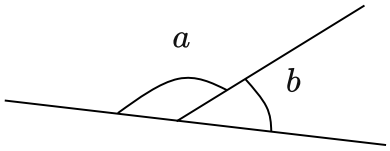
$$a^2 = b^2 + c^2 - 2bc\cos A$$

The idea is that we can use the common length of ΔACX and $\angle XBC$, i.e. h, to connect the two triangles, and therefore connect their angles/length.

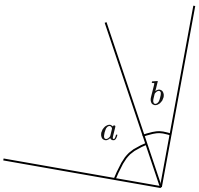


Using basic trigonometry:
$h = a\sin B$ and $h = b\sin A$
$\therefore a\sin B = b\sin A$
$\therefore \frac{a}{\sin A} = \frac{b}{\sin B}$

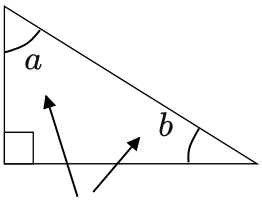
Why does "cosine" contains the word "sine"?



Supplementary Angles
add to 180°

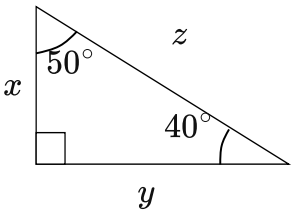


Complementary Angles
add to 90°



Therefore these angles are complementary.

i.e. The **cosine** of an angle is the **sine** of the **complementary** angle.
Hence **cosine = COMPLEMENTARY SINE**



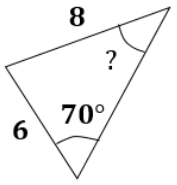
$\cos(50) =$ ☐

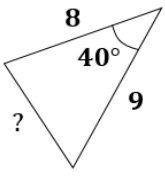
$\sin(40) =$ ☐

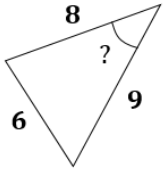
$\therefore \cos(50) = \sin(40)$

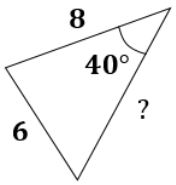
When triangles are not right-angled, we can no longer use simple trigonometric ratios, and must use the cosine and sine rules.

You have	You want	
#1: Two angle-side opposite pairs	Missing angle or side in one pair	
#2 Two sides known and a missing side opposite a known angle	Remaining side	
#3 All three sides	An angle	
#4 Two sides known and a missing side not opposite known angle	Remaining side	

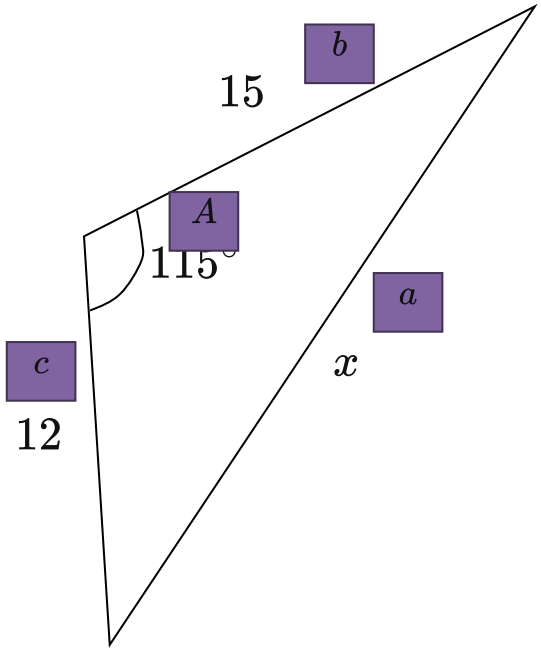








We use the **cosine rule** whenever we have **three sides** (and an angle) involved.



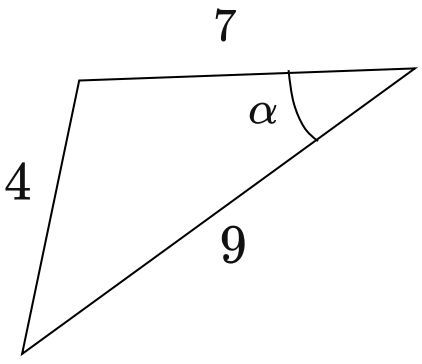
Proof at end of PowerPoint.

Cosine Rule:

$$a^2 = b^2 + c^2 - 2bc\cos A$$

You have	You want	Use
All three sides	An angle	Cosine rule

$$\mathbf{a^2 = b^2 + c^2 - 2bc\cos A}$$



Tip: The brackets are not needed, but students who forget about BIDMAS see $4^2 = (7^2 + 9^2 - 2 \times 7 \times 9)\cos\alpha$ and hence incorrectly simplify to $16 = 4\cos\alpha$

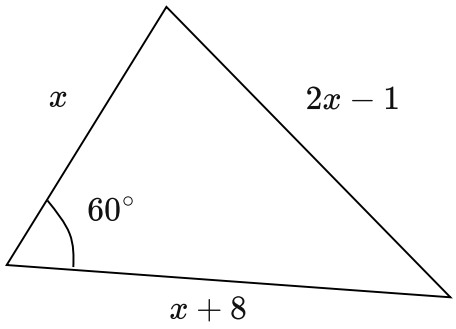
Label sides then substitute into formula.

Simplify each bit of formula.

Rearrange "swapping" thing you're subtracting and result

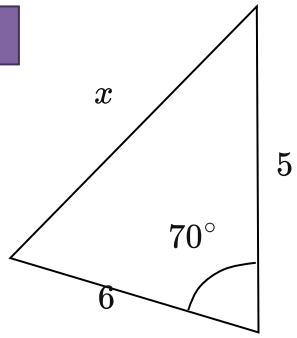
Textbook Note: The textbook presents the rearrangement of the cosine rule: $\cos A = \frac{b^2+c^2-a^2}{2bc}$ to find missing angles. It is not advisable to memorise this

Determine the value of x.



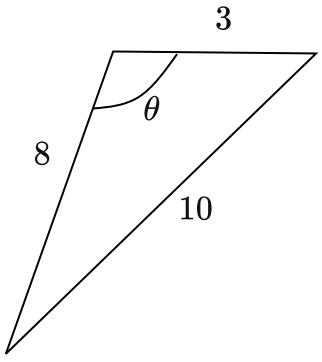
[From textbook] Coastguard station B is 8 km, on a bearing of 060°, from

1



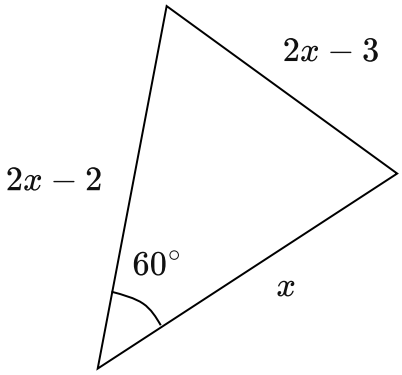
$x =$

2



$\theta =$

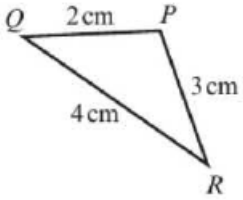
3



$x =$

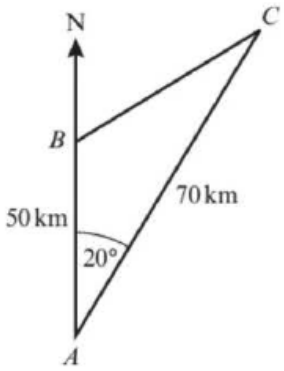
Note: You will get an obtuse angle whenever you inverse cos of a negative value.

8 Show that $\cos P = -\frac{1}{4}$.



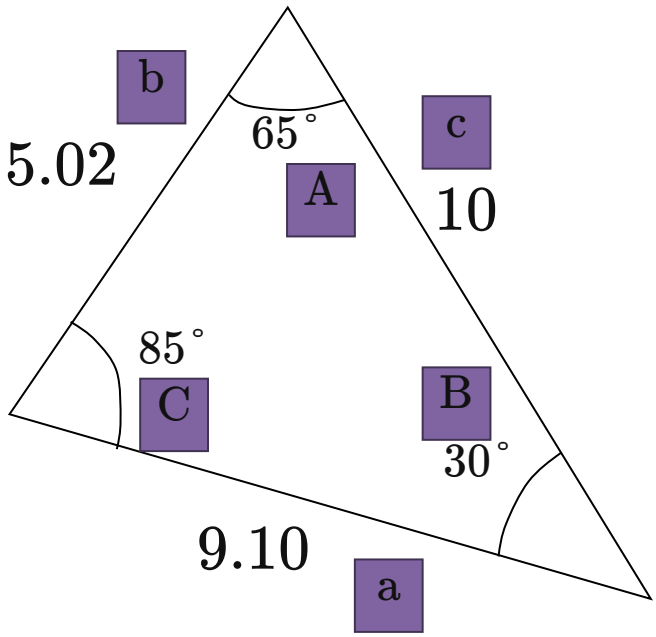
11 The lengths of the sides of a triangle are in the ratio 2 : 3 : 4.
Calculate the size of the largest angle.

17 The diagram shows three cargo ships, A, B and C, which are in the same horizontal plane. Ship B is 50 km due north of ship A and ship C is 70 km from ship A. The bearing of C from A is 020°.
a Calculate the distance between ships B and C, in kilometres to 3 s.f. (3 marks)
b Calculate the bearing of ship C from ship B. (4 marks)



1 *[STEP I 2009 Q4i]*
The sides of a triangle have lengths $p-q$, p and $p+q$, where $p > q > 0$. The largest and smallest angles of the triangle are α and β respectively. Show by means of the cosine rule that

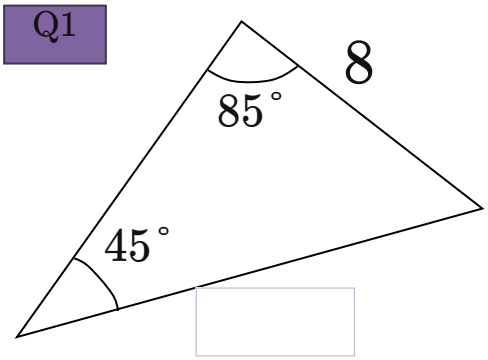
$$4\,(1-\cos\alpha)\,(1-\cos\beta) = \cos\alpha + \cos\beta$$



For this triangle, try calculating each side divided by the sin of its opposite angle. What do you notice in all three cases?

You have	You want	Use
#1: Two angle-side opposite pairs	Missing angle or side in one pair	Sine rule

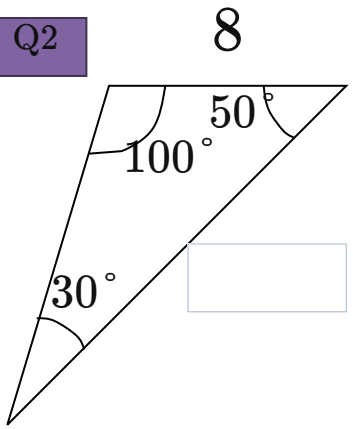
Q1



$$\frac{x}{\sin 85} = \frac{8}{\sin 45}$$

$$x = \frac{8\sin 85}{\sin 45} = 11.27$$

Q2



$$\frac{x}{\sin 100} = \frac{8}{\sin 30}$$

$$x = \frac{8\sin 100}{\sin 30} = 15.76$$

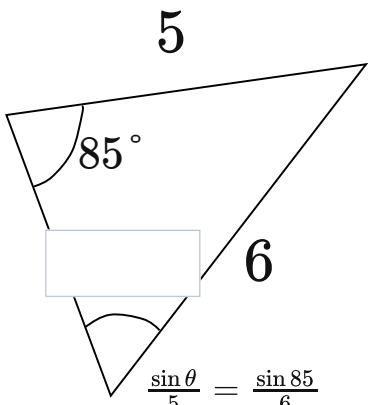
You have	You want	Use
#1: Two angle-side opposite pairs	Missing angle or side in one pair	Sine rule

When you have a missing angle, it's better to reciprocate to get:

$$\frac{\sin A}{a} = \frac{\sin B}{b}$$

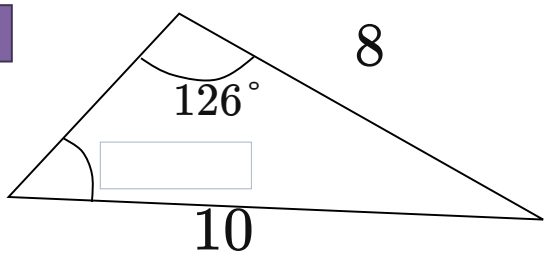
i.e. in general put the missing value in the numerator.

Q3



$$\frac{\sin\theta}{5} = \frac{\sin 85}{6}$$
$$\sin\theta = \frac{5\sin 85}{6}$$
$$\theta = \sin^{-1}\left(\frac{5\sin 85}{6}\right)$$
$$= 56.11^\circ$$

Q4



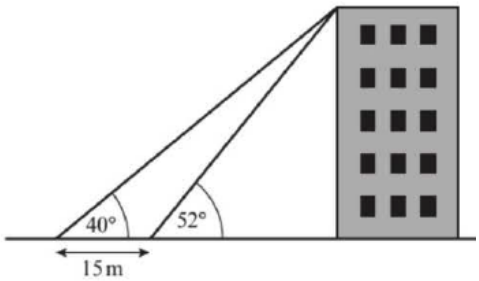
$$\frac{\sin\theta}{8} = \frac{\sin 126^\circ}{10}$$
$$\sin\theta = \frac{8\sin 126}{10}$$
$$\theta = \sin^{-1}\left(\frac{8\sin 126}{10}\right)$$
$$= 40.33^\circ$$

8 Town B is 6 km, on a bearing of 020°, from town A. Town C is located on a bearing of 055° from town A and on a bearing of 120° from town B. Work out the distance of town C from:
a town A $btown B$

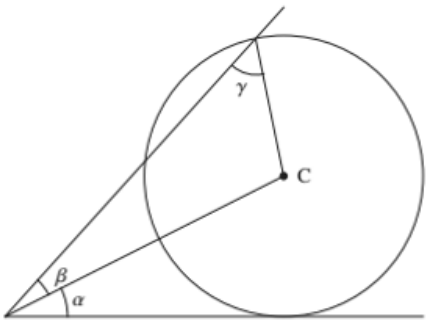
11 In $\triangle ABC$, $AB = x$ cm, $BC = (4 - x)$ cm, $\angle BAC = y$ and $\angle BCA = 30^\circ$.
Given that $\sin y = \frac{1}{\sqrt{2}}$, show that

$$x = 4(\sqrt{2} - 1)$$

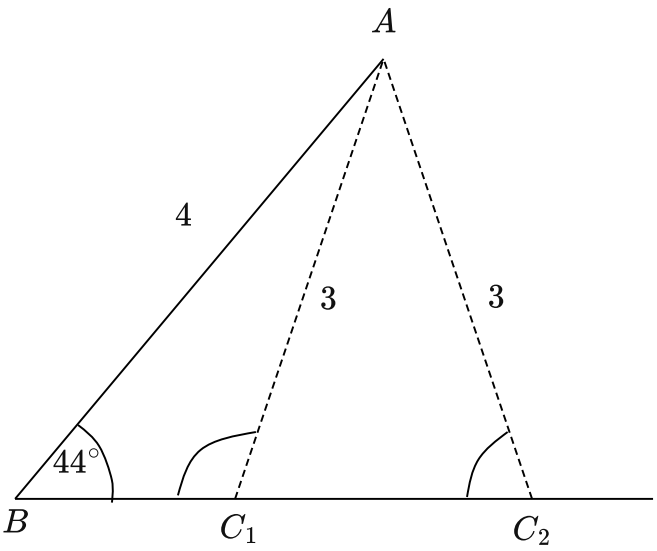
12 A surveyor wants to determine the height of a building. She measures the angle of elevation of the top of the building at two points 15 m apart on the ground.
a Use this information to determine the height of the building. (4 marks)
b State one assumption made by the surveyor in using this mathematical model. (1 mark)



1 *[MAT 2011 1E]*
The circle in the diagram has centre C. Three angles α, β, γ are also indicated.



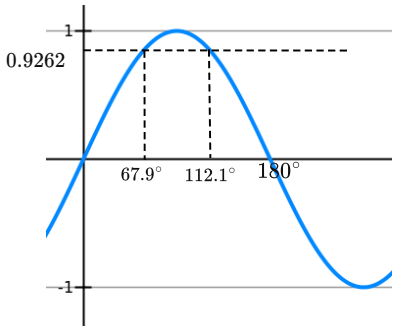
T equation:
A) $\cos\alpha = \sin(\beta+\gamma)$
B) $\sin\beta = \sin\alpha\sin\gamma$
C) $\sin\beta(1-\cos\alpha) = \sin\gamma$
D) $\sin(\alpha+\beta) = \cos\gamma\sin\alpha$



Suppose you are told that $AB = 4$, $AC = 3$ and $\angle ABC = 44°$. What are the possible values of $\angle ACB$?

C is somewhere on the horizontal line. There's two ways in which the length could be 3. Using the sine rule:

$$\frac{\sin C}{4} = \frac{\sin 44}{3}$$
$$C = \sin^{-1}(0.9262)$$

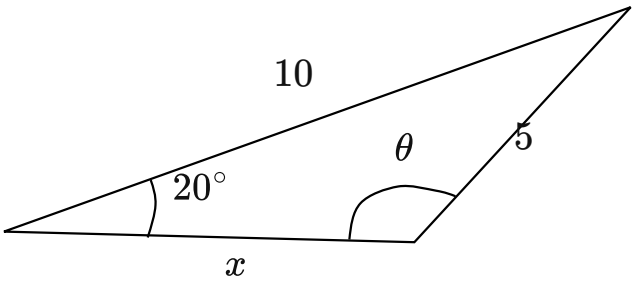


Your calculator will give the acute angle of $67.9°$ (i.e. C_2). But if we look at a graph of sin, we can see there's actually a second value for $\sin^{-1}(0.9262)$, corresponding to angle C_1.

✎ The sine rule produces two possible solutions for a missing angle:

$$\sin\theta = \sin(180° - \theta)$$

Whether we use the acute or obtuse angle depends on context.

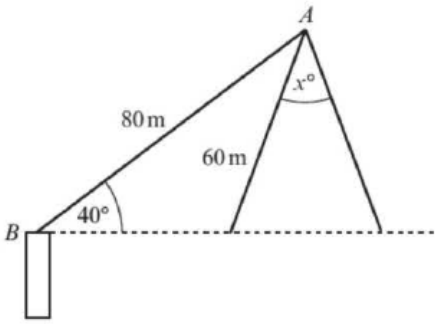


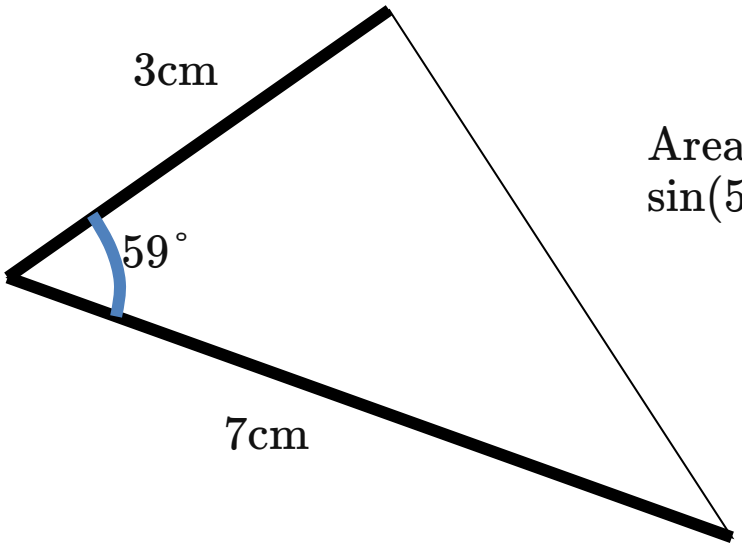
Given that the angle θ is obtuse, determine θ and hence determine the length of x.

4 Triangle ABC is such that $AB = 4$ cm, $BC = 6$ cm and $\angle ACB = 36°$. Show that one of the possible values of $\angle ABC$ is $25.8°$ (to 3 s.f.). Using this value, calculate the length of AC.

5 Two triangles ABC are such that $AB = 4.5$ cm, $BC = 6.8$ cm and $\angle ACB = 30°$. Work out the value of the largest angle in each of the triangles.

6 a A crane arm AB of length 80 m is anchored at point B at an angle of 40° to the horizontal. A wrecking ball is suspended on a cable of length 60 m from A. Find the angle x through which the wrecking ball rotates as it passes the two points level with the base of the crane arm at B. (6 marks)
b Write down one modelling assumption you have made. (1 mark)



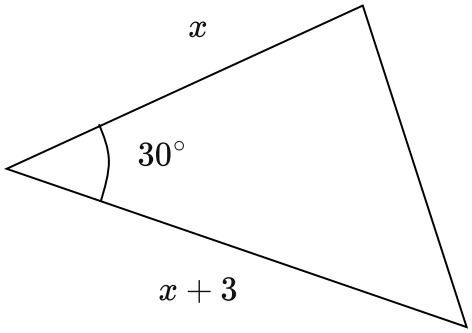


Area
sin(5

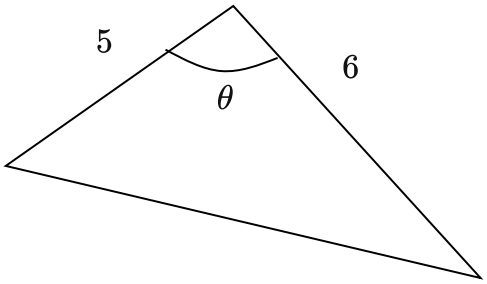
$$\text{Area} = \frac{1}{2}ab\sin(C)$$

where C is the angle between two sides a and b.

Tip: You shouldn't have to label sides/angles before using the formula. Just remember that the angle is between the two sides

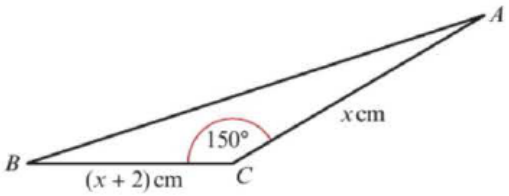


The area of this triangle is 10.
Determine x.



The area of this triangle is also 10.
If θ is obtuse, determine θ.

4 In triangle
ABC, $BC = (x + 2)$ cm, $AC = x$ cm and
$\angle BCA = 150^\circ$. Given that the area of the triangle is 5 cm^2, work out the value of x, giving your answer to 3 significant figures.



5 In $\triangle PQR$, $PQ = (x + 2)$ cm, $PR = (5 - x)$ cm and $\angle QPR = 30°$. The area of the triangle is A cm^2.

a Show that $A = \frac{1}{4}(10 + 3x - x^2)$.

b Use the method of completing the square, or otherwise, to find the maximum value of A, and give the corresponding value of x.

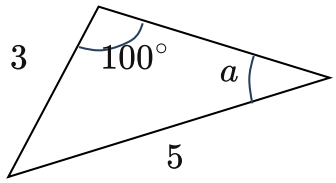
6 In $\triangle ABC$, $AB = x$ cm, $AC = (5 + x)$ cm and $\angle I$

Given that the area of the triangle is $3\frac{3}{4}$ cm^2

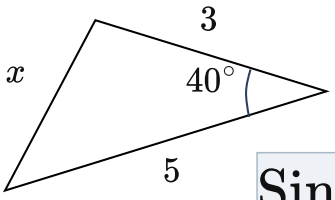
a Show that x satisfies the equation $x^2 + 5x - 15$

b Calculate the value of x, giving your answer to 3 significant figures.

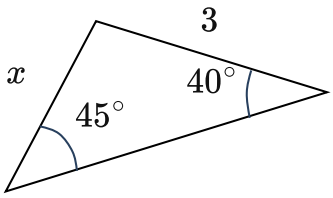
Sine Cosine



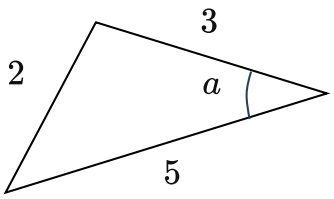
Recall that whenever we have **two "side-angle pairs"** involved, use sine rule. If there's **3 sides** involved, we can use cosine rule. Sine rule is generally easier to use than cosine rule.



Sine Cosine

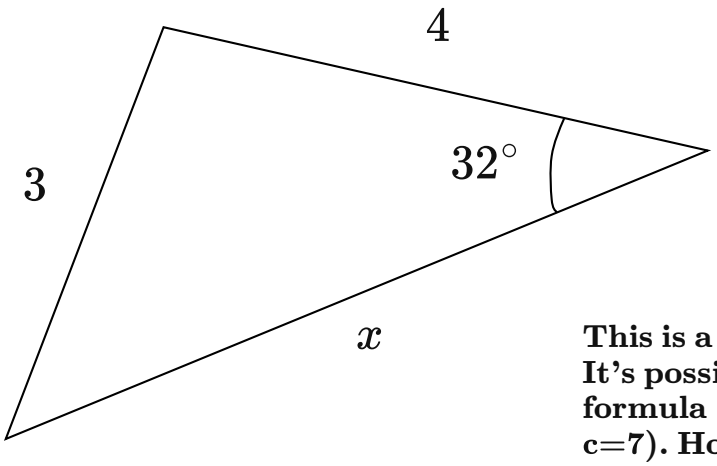


Sine Cosine



Sine Cosine

You have	You want	Use
#4 Two sides known and a missing side not opposite known angle	Remaining side	Sine rule twice

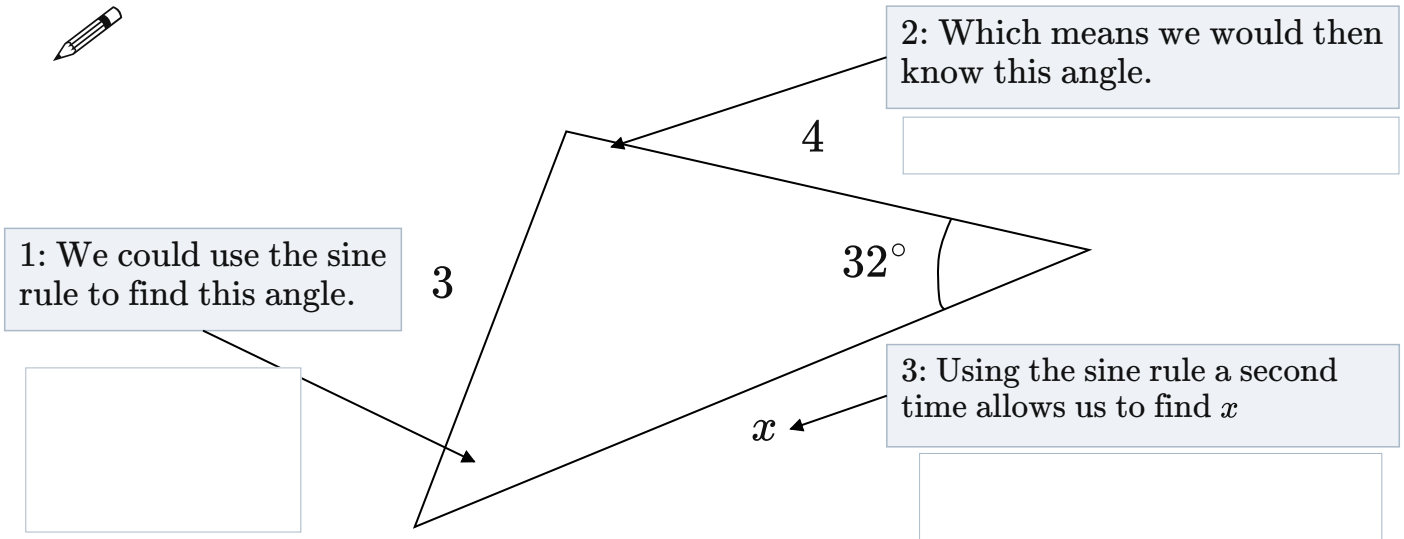


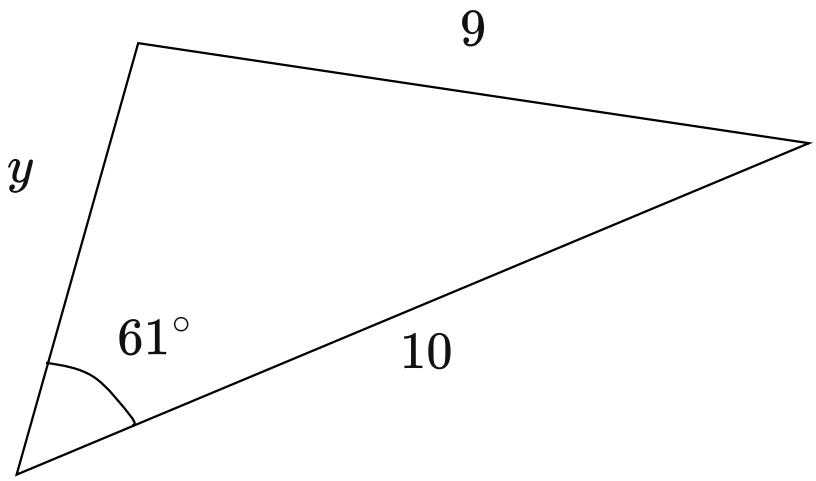
Given there is just one angle involved, you might attempt to use the cosine rule:

$$\mathbf{3^2 = x^2 + 4^2 - (2 \times x \times 4 \times \cos 32)}$$

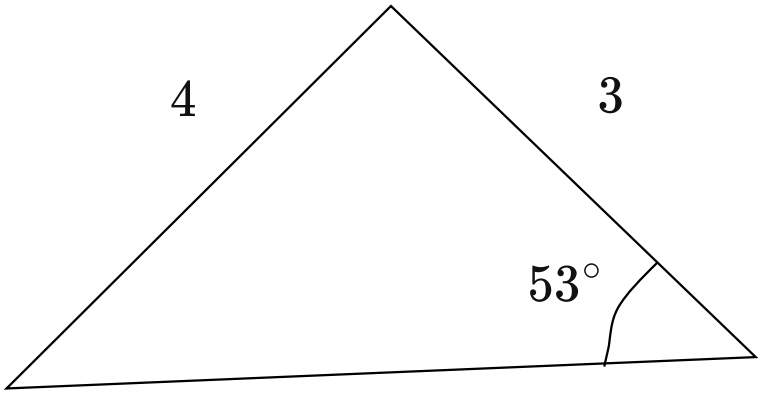
This is a quadratic equation!
It's possible to solve this using the quadratic formula (using $a=1, b=-8\cos 32$, $c=7$). However, this is a bit fiddly and not the primary method expected in the exam...

You have	You want	Use
#4 Two sides known and a missing side not opposite known angle	Remaining side	Sine rule twice





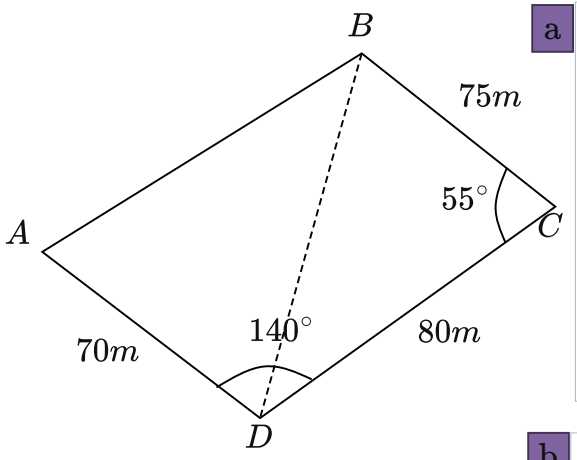
$y =$



$Area =$

[From Textbook] The diagram shows the locations of four mobile phone masts in a field, $BC = 75m$.
$CD = 80m$, angle $BCD = 55°$ and angle $ADC = 140°$.
In order that the masts do not interfere with each other, they must be at least 70m apart.
Given that A is the minimum distance from D, find:

a) The distance A is from B
b) The angle BAD
c) The area enclosed by the four masts.



a

By adding areas of ΔABD and ΔBDC:

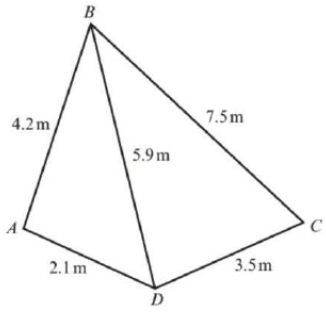
b

c

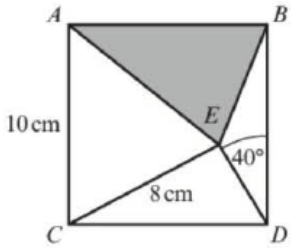
12 The perimeter of $\triangle ABC = 15$ cm. Given that $AB = 7$ cm and $\angle BAC = 60°$, find the lengths of AC and BC and the area of the triangle.

13 In the triangle ABC, $AB = 14$ cm, $BC = 12$ cm and $CA = 15$ cm.
a Find the size of angle C, giving your answer to 3 s.f.
b Find the area of triangle ABC, giving your answer in cm^2 to 3 s.f.

14 A flower bed is in the shape of a quadrilateral as shown in the diagram.
a Find the sizes of angles DAB and BCD. (4 ma . . .
b Find the total area of the flower bed. (3 ma . . .
c Find the length of the diagonal AC.



15 $ABCD$ is a square. Angle CED is obtuse.
Find the area of the shaded triangle. (7 marks)



[AEA 2009 Q5a] The sides of the triangle ABC have lengths $BC = a$, $AC = b$ and $AB = c$, where $a < b < c$. The sizes of the angles A, B and C form an arithmetic sequence.

(i) Show that the area of triangle ABC is $ac\frac{\sqrt{3}}{4}$.

Given that $a = 2$ and $\sin A = \frac{\sqrt{15}}{5}$, find

(ii) the value of b,

(iii) the value of c.

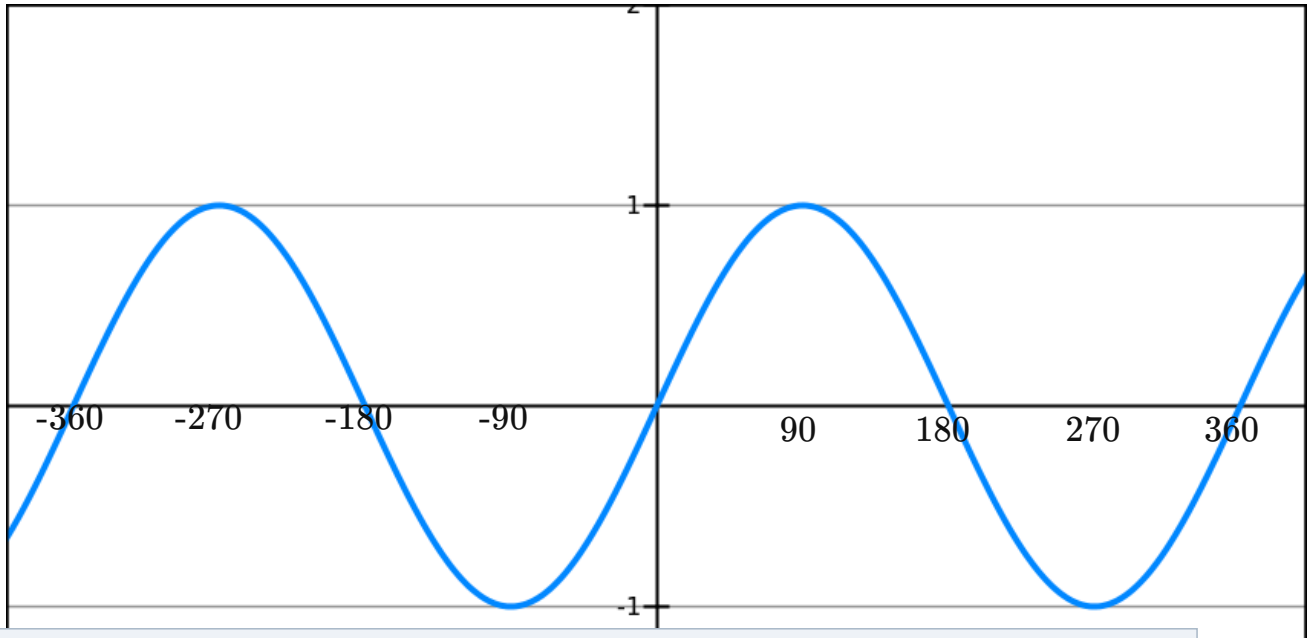
[STEP I 2006 Q8] Note that the volume of a tetrahedron is equal to
$\frac{1}{3} \times areaofbase \times height$
The points O, A, B, C have coordinates $(0,0,0), (a,0,0), (0,b,0)$ and $(0,0,c)$ respectively, where a, b, c are positive.

(i) Find, in terms of a, b, c the volume of the tetrahedron $OABC$.

(ii) Let angle $ACB = \theta$. Show that $\cos\theta = \frac{c^2}{\sqrt{(a^2+c^2)(b^2+c^2)}}$ and find, in terms of a, b and c, the area of triangle ABC.
Hence show that d, the perpendicular distance of the origin from the triangle ABC, satisfies $\frac{1}{d^2} = \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}$

What does it look like?

What do the following graphs look like?

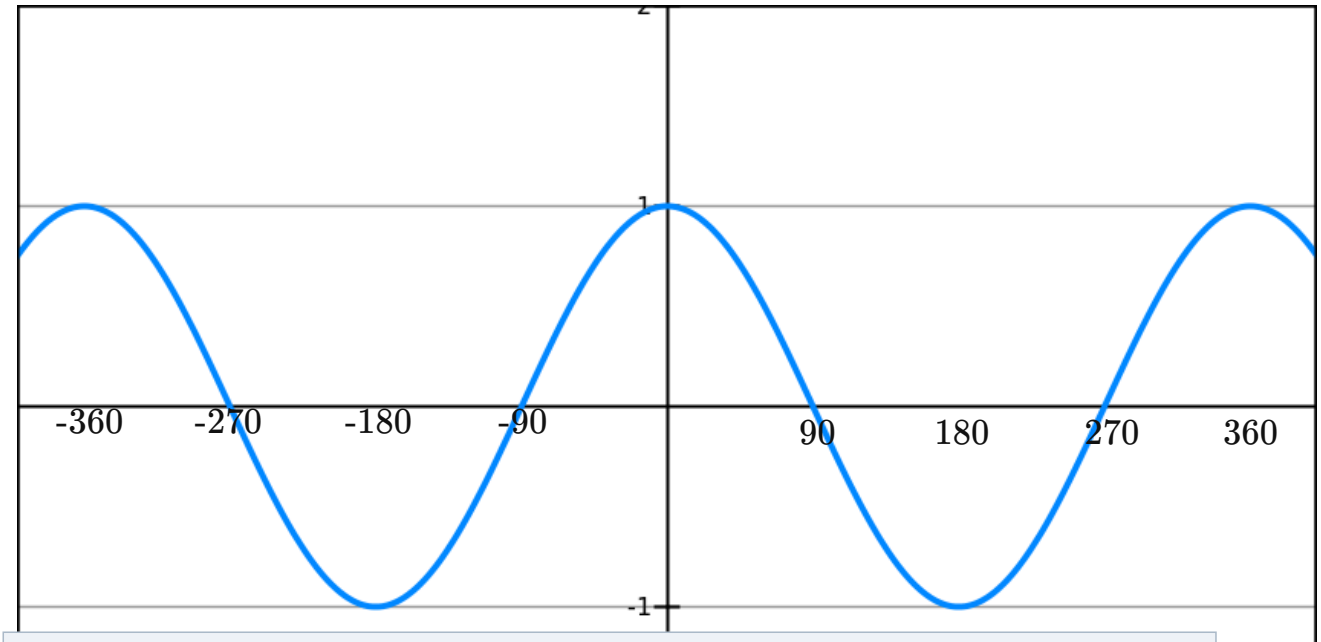


Suppose we know that **sin(30) = 0.5**. By thinking about symmetry in the graph, how could we work out:

sin(150) sin(-30) sin(210)

What do the following graphs look like?

What does it look like?

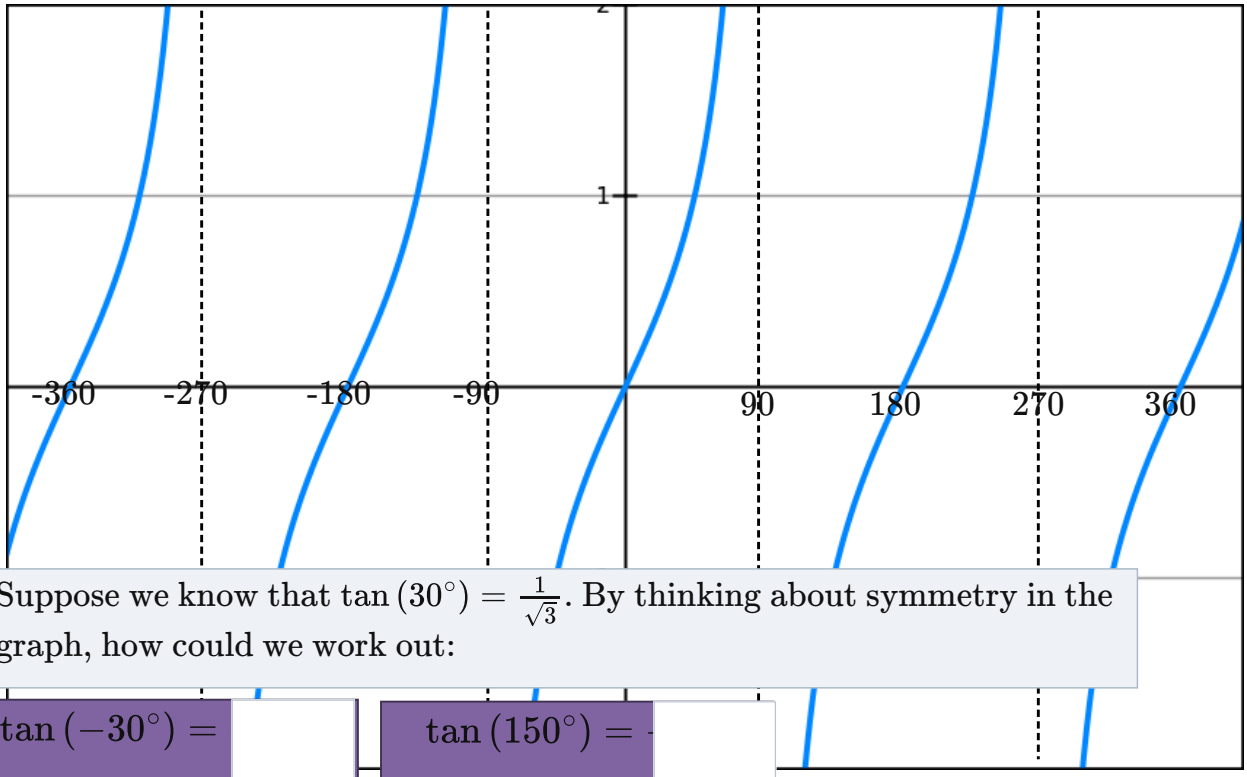


Suppose we know that **cos(60) = 0.5**. By thinking about symmetry in the graph, how could we work out:

cos(120) cos(-60) cos(240)

What does it look like?

What does it look like?



Suppose we know that $\tan(30°) = \frac{1}{\sqrt{3}}$. By thinking about symmetry in the graph, how could we work out:

$\tan(-30°) =$

$\tan(150°) =$

There is no new theory here: just use your knowledge of transforming graphs, i.e. whether the transformation occurs 'inside' the function (i.e. input modified) or 'outside' the function (i.e. output modified).

Sketch $y = 4\sin x$, $0 \leq x \leq 360°$

Sketch $y = \cos(x + 45°)$, $0 \leq x \leq 360°$

Sketch $y = -\tan x, 0 \leq x \leq 360^\circ$

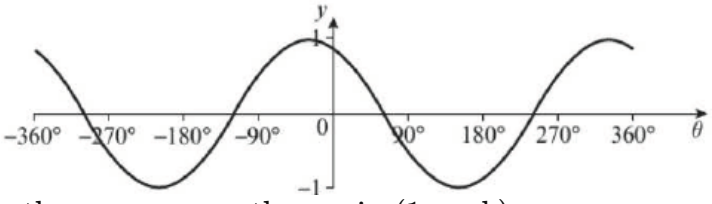
Sketch $y = \sin\left(\frac{x}{2}\right), 0 \leq x \leq 360^\circ$

Exercise 9F – Y1 Page 194 52

1 Sketch the graph of $y = \cos\theta$ in the interval $-180^\circ \leq \theta \leq 180^\circ$.
2 Sketch the graph of $y = \tan\theta$ in the interval $-180^\circ \leq \theta \leq 180^\circ$.
3 Sketch the graph of $y = \sin\theta$ in the interval $-90^\circ \leq \theta \leq 270^\circ$.

7 The graph shows the curve $y = \cos(x + 30^\circ), -360^\circ \leq x \leq 360^\circ$.

a Write down the coordinates of the points where the curve crosses the x-axis. (2 marks)



b Find the coordinates of the point where the curve crosses the y-axis. (1 mark)

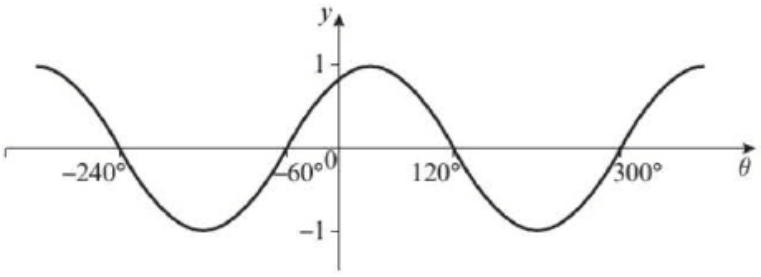
b

$\cos 30^\circ = \frac{\sqrt{3}}{2}$; $\left(0, \frac{\sqrt{3}}{2}\right)$

8 The graph shows the curve with equation $y = \sin(x + k), -360^\circ \leq x \leq 360^\circ$, where k is a constant.

a Find one possible value for k. (2 marks)

b Is there more than one possible answer to part a? Give a reason for your answer. (2 marks)



9 The variation in the depth of water in a rock pool can be modelled using the function $y = \sin(30t)^\circ$, where t is the time in hours and $0 \leq t \leq 6$.
a Sketch the function for the given interval.
b If $t = 0$ represents midday, during what times will the rock pool be at least half full?

1 *[MAT 2013 1B]* The graph of $y = \sin x$ is reflected first in the line $x = \pi$ and then in the line $y = 2$. The resulting graph has equation:

A) $y = \cos x$
B) $y = 2 + \sin x$
C) $y = 4 + \sin x$
D) $y = 2 - \cos x$

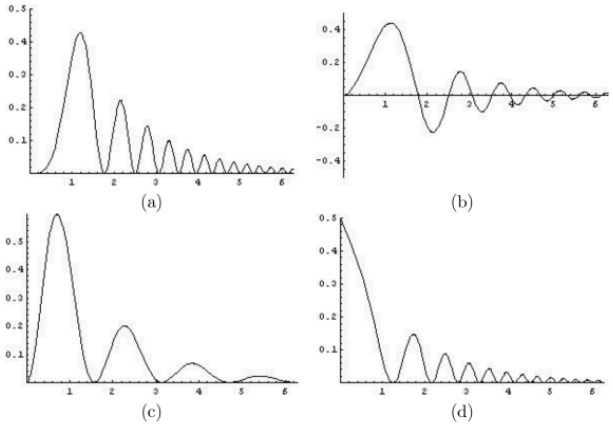
Solution: C

2 *[MAT 2011 1D]* What fraction of the interval $0 \leq x \leq 360^\circ$ is one (or both) of the inequalities: $\sin x \geq \frac{1}{2}$, $\sin 2x \geq \frac{1}{2}$ true?

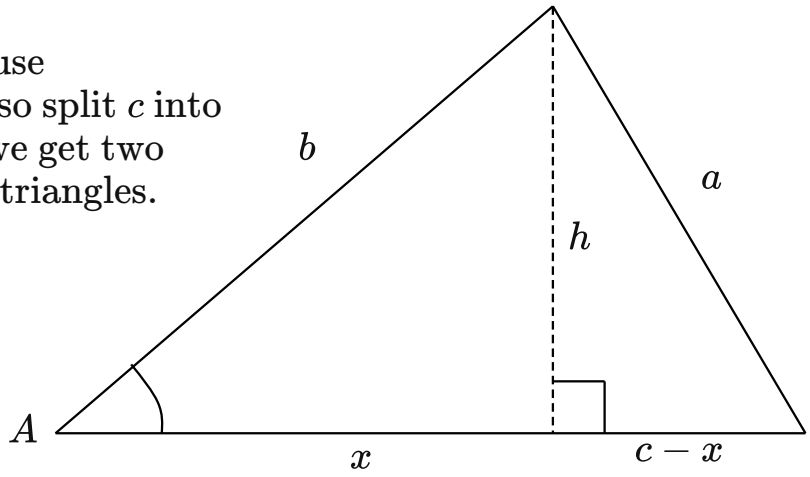
Solution: $\frac{13}{24}$ (this is clear if you draw the)

3 *[MAT 2007 1G]* On which of the axes is a sketch of the graph

$$y = 2^{-x}\sin^2\left(x^2\right)$$



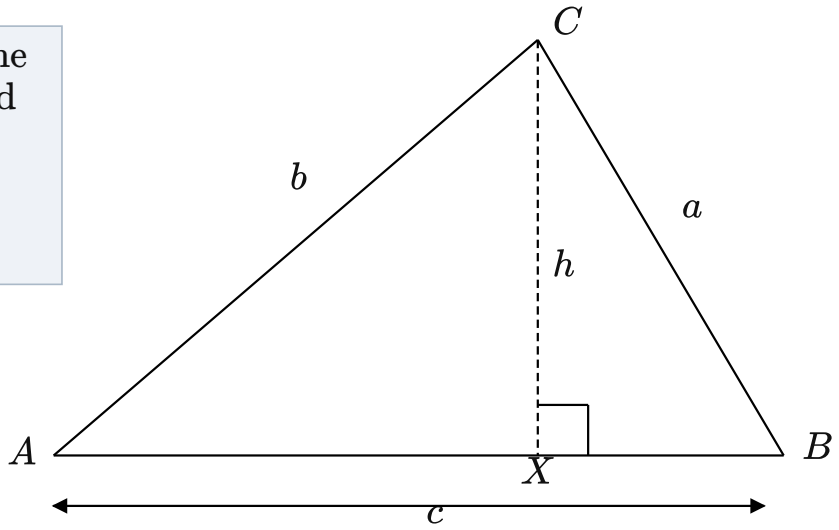
We want to use Pythagoras, so split c into two so that we get two right-angled triangles.



By Pythagoras: $h^2 + x^2 = b^2$ and $h^2 + (c-x)^2 = a^2$
Subtracting to eliminate h:
$x^2 - (c-x)^2 = b^2 - a^2$
$a^2 = b^2 + c^2 - 2cx$
But $x = b\cos A$ $\therefore$

$$a^2 = b^2 + c^2 - 2bc\cos A$$

The idea is that we can use the common length of ΔACX and $\angle XBC$, i.e. h, to connect the two triangles, and therefore connect their angles/length.



Using basic trigonometry:
$h = a\sin B$ and $h = b\sin A$
$\therefore a\sin B = b\sin A$
$\therefore \frac{a}{\sin A} = \frac{b}{\sin B}$